



Assignment -04

Course Details

Course Title: Data Communications

Couse Code: CSE 350

Section: 01

Submitted To

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Introduction

In this assignment, I design, configure, and verify a multi router network using only Class B IP addressing. In the real world, networks are often divided into multiple subnets connected through routers to ensure efficient routing, scalability and security.

Here, I implement a network consisting of three routers, each connected to its own LAN. The routers are interconnected via serial DCE links, forming a triangular topology that provides multiple routing paths between networks. Static routing is used to manually define routes to each destination network, which helps in understanding routing mechanisms and packet flow across an internetwork.

Network Topology Overview

The topology consists of

- 3 Routers (Router0, Router1, Router2)
- 3 LAN networks, each containing 2 PCs and 1 switch
- 3 serial DCE connections between routers
- Class B IP addressing only

IP Addressing

Device	Interface	IP Address	Subnet Mask	Network
PC0	Fa0	169.100.0.1	255.255.0.0	169.100.0.0
PC1	Fa0	169.100.0.2	255.255.0.0	169.100.0.0
Router0	Fa0/0	169.100.0.254	255.255.0.0	169.100.0.0
Router0	S2/0	139.100.0.1	255.255.0.0	139.100.0.0
Router0	S3/0	149.100.0.1	255.255.0.0	149.100.0.0
PC2	Fa0	179.100.0.1	255.255.0.0	179.100.0.0
PC3	Fa0	179.100.0.2	255.255.0.0	179.100.0.0
Router1	Fa0/0	179.100.0.254	255.255.0.0	179.100.0.0
Router1	S2/0	149.100.0.2	255.255.0.0	149.100.0.0
Router1	S3/0	159.100.0.1	255.255.0.0	159.100.0.0
PC4	Fa0	189.100.0.1	255.255.0.0	189.100.0.0
PC5	Fa0	189.100.0.2	255.255.0.0	189.100.0.0
Router2	Fa0/0	189.100.0.254	255.255.0.0	189.100.0.0
Router2	S2/0	139.100.0.2	255.255.0.0	139.100.0.0
Router2	S3/0	159.100.0.2	255.255.0.0	159.100.0.0

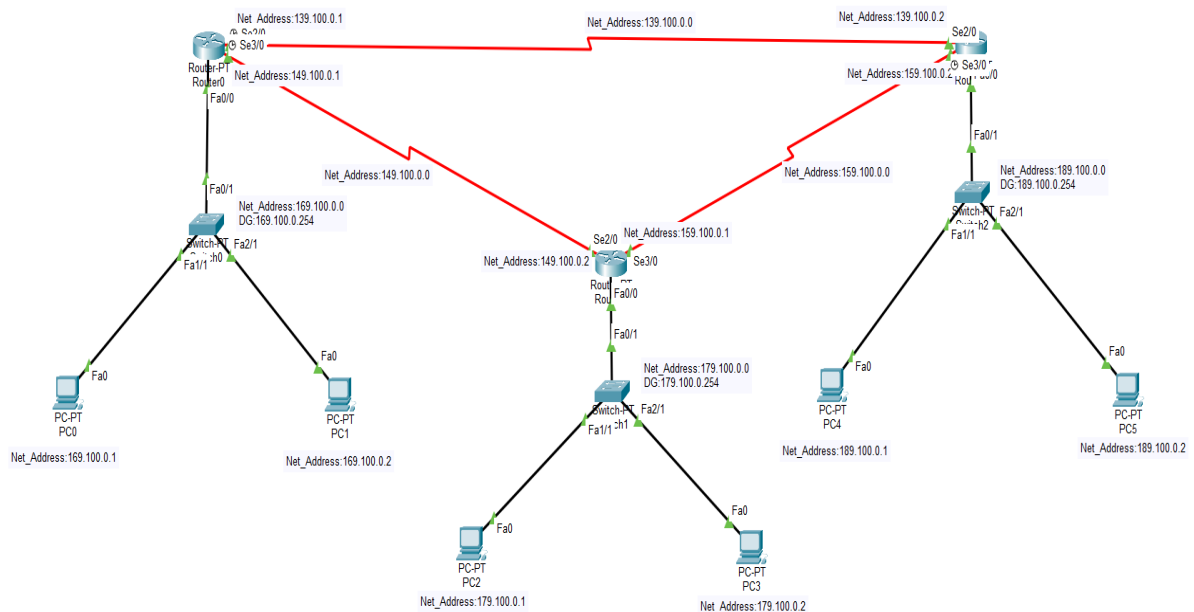


Figure: Network diagram with Static Router Protocol

Router Configurations

For Router 0:

Se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 139.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 139.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 149.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se3/0
Router(config-if)#ip address 149.100.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

fa0/0

```
Router(config)# int fa0/0
Router(config-if)#ip address 169.100.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)# int fa0/0
Router(config-if)#ip address 169.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

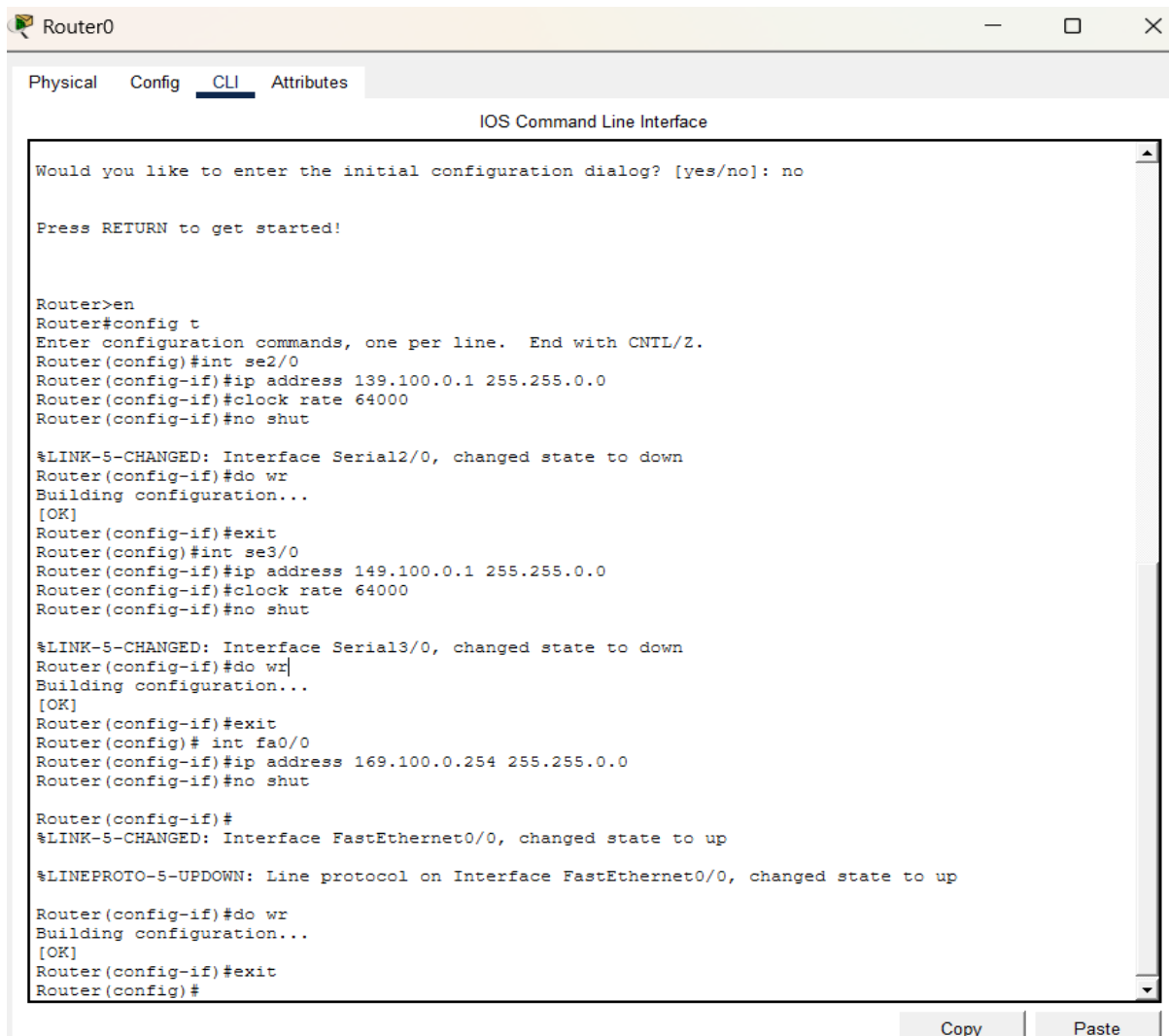


Figure: Router0 configuration

Router0 (Static Configuration)

Router>en

Router#config

Router(config)#ip route 159.100.0.0 255.255.0.0 149.100.0.2

Router(config)#ip route 179.100.0.0 255.255.0.0 149.100.0.2

Router(config)#ip route 189.100.0.0 255.255.0.0 149.100.0.2

Router(config)#exit

```

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 159.100.0.0 255.255.0.0 149.100.0.2
Router(config)#ip route 179.100.0.0 255.255.0.0 149.100.0.2
Router(config)#ip route 189.100.0.0 255.255.0.0 149.100.0.2
Router(config)#exit

```

For Router 1

se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 149.100.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 149.100.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
Router(config-if)#exit
_
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 159.100.0.1 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se3/0
Router(config-if)#ip address 159.100.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

fa0/0

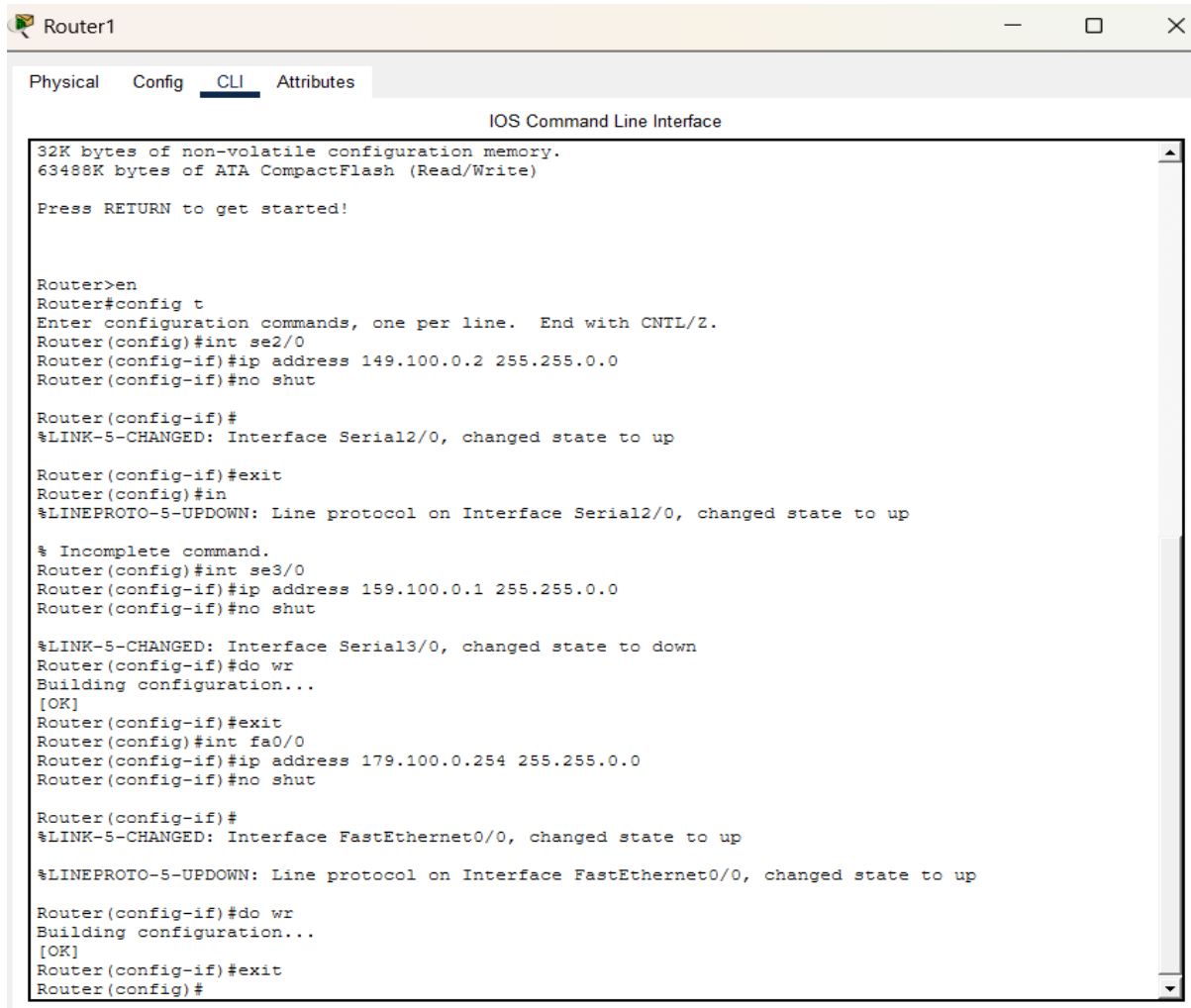
```
Router(config)#int fa0/0
Router(config-if)#ip address 179.100.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa0/0
Router(config-if)#ip address 179.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```



The screenshot shows a window titled "Router1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 149.100.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#in
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

% Incomplete command.
Router(config)#int se3/0
Router(config-if)#ip address 159.100.0.1 255.255.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 179.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Figure: Router1 configuration

Router1 (Static Configuration)

```
Router(config)#ip route 139.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 169.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 189.100.0.0 255.255.0.0 159.100.0.2
Router(config)#exit
```

```
Router(config)#ip route 139.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 169.100.0.0 255.255.0.0 159.100.0.2
Router(config)#ip route 189.100.0.0 255.255.0.0 159.100.0.2
Router(config)#exit
```

For Router 2

se2/0

```
Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 139.100.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 139.100.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#
```

Se3/0

```
Router(config)#int se 3/0
Router(config-if)#ip address 159.100.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se 3/0
Router(config-if)#ip address 159.100.0.2 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 189.100.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```



```

Router(config)#int fa0/0
Router(config-if)#ip address 189.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

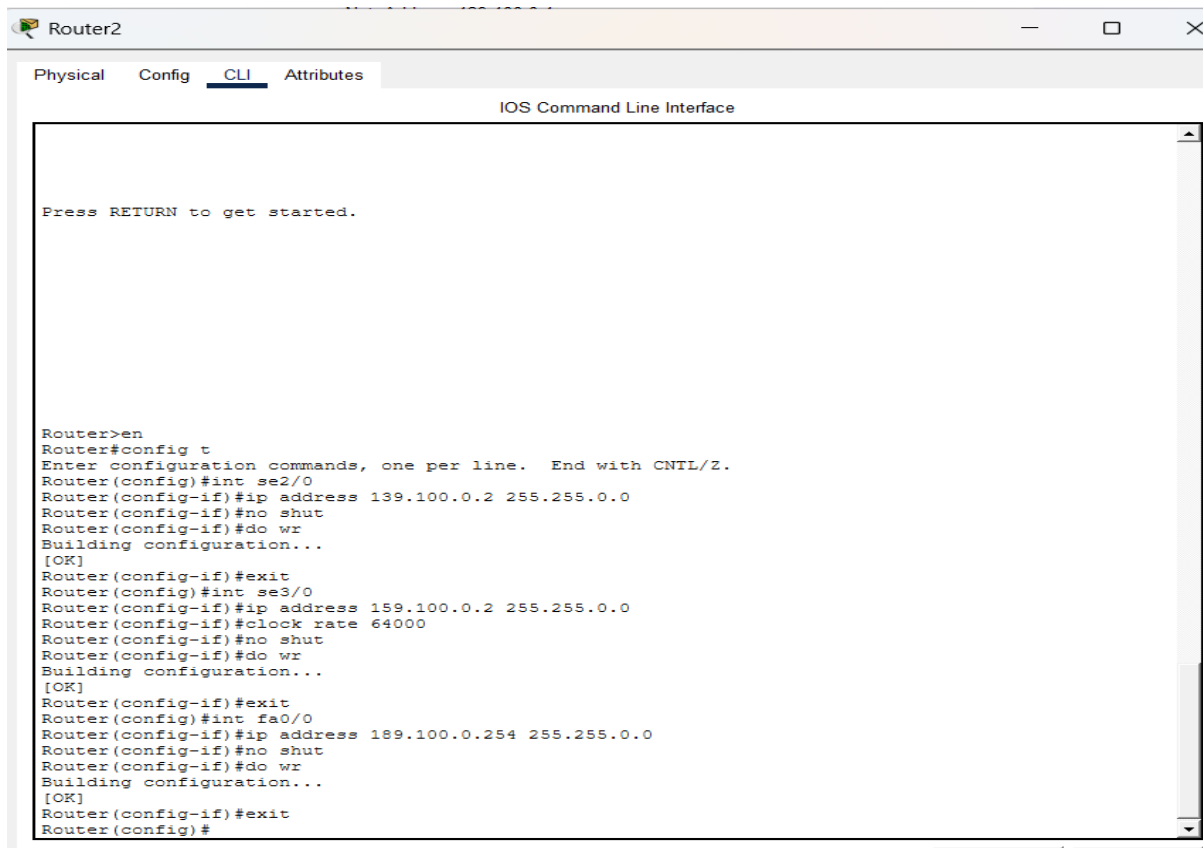


Figure: Router2 configuration

Router2 (Static Configuration)

```

Router(config)#ip route 149.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 169.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 179.100.0.0 255.255.0.0 139.100.0.1
Router(config)#exit

```

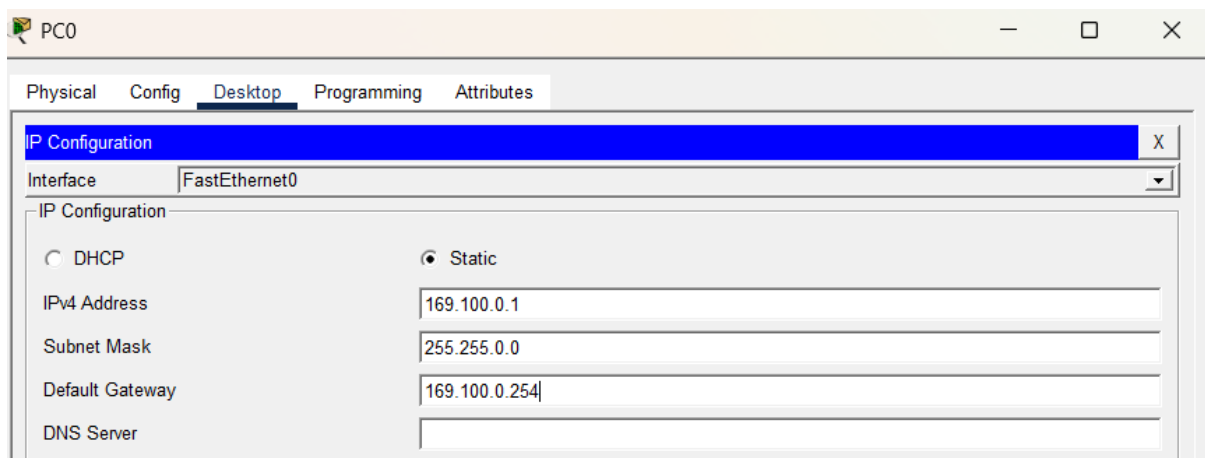
```

Router(config)#ip route 149.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 169.100.0.0 255.255.0.0 139.100.0.1
Router(config)#ip route 179.100.0.0 255.255.0.0 139.100.0.1
Router(config)#exit

```

Device Configuration

For PC 0

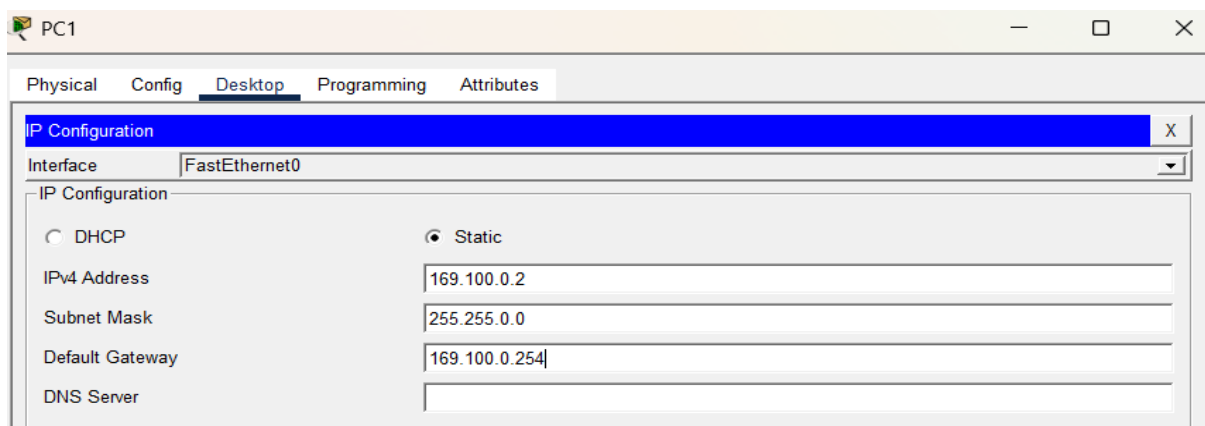


The screenshot shows the configuration window for PC0. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The fields are filled with the following values:

Field	Value
IPv4 Address	169.100.0.1
Subnet Mask	255.255.0.0
Default Gateway	169.100.0.254
DNS Server	

Figure: PC Configuration of PC0

For PC 1

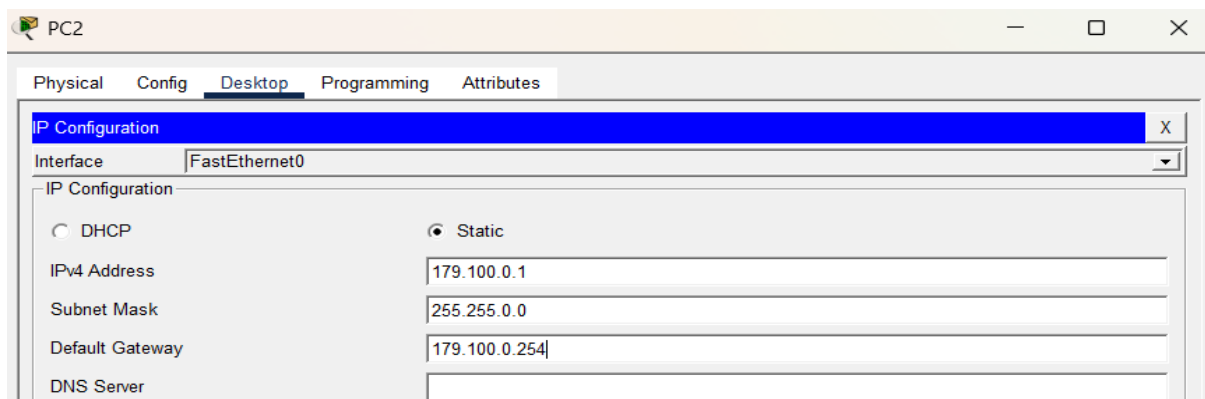


The screenshot shows the configuration window for PC1. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The fields are filled with the following values:

Field	Value
IPv4 Address	169.100.0.2
Subnet Mask	255.255.0.0
Default Gateway	169.100.0.254
DNS Server	

Figure: PC Configuration of PC1

For PC 2



The screenshot shows the configuration window for PC2. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The fields are filled with the following values:

Field	Value
IPv4 Address	179.100.0.1
Subnet Mask	255.255.0.0
Default Gateway	179.100.0.254
DNS Server	

Figure: PC Configuration of PC2

For PC 3

The screenshot shows the configuration window for PC3. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Interface' is 'FastEthernet0'. The 'Static' radio button is selected. The fields are filled with: IPv4 Address: 179.100.0.2, Subnet Mask: 255.255.0.0, Default Gateway: 179.100.0.254, and DNS Server is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	179.100.0.2
Subnet Mask	255.255.0.0
Default Gateway	179.100.0.254
DNS Server	

Figure: PC Configuration of PC3

For PC 4

The screenshot shows the configuration window for PC4. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Interface' is 'FastEthernet0'. The 'Static' radio button is selected. The fields are filled with: IPv4 Address: 189.100.0.1, Subnet Mask: 255.255.0.0, Default Gateway: 189.100.0.254, and DNS Server is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	189.100.0.1
Subnet Mask	255.255.0.0
Default Gateway	189.100.0.254
DNS Server	

Figure: PC Configuration of PC4

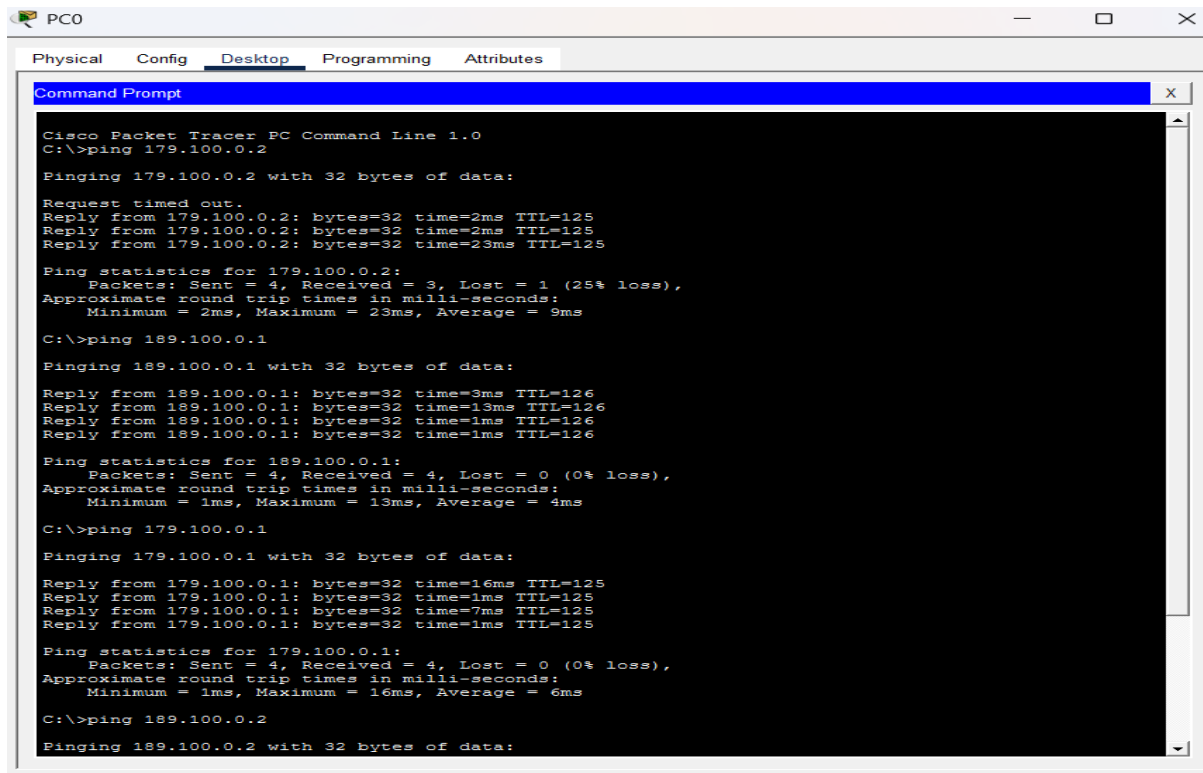
For PC5

The screenshot shows the configuration window for PC5. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Interface' is 'FastEthernet0'. The 'Static' radio button is selected. The fields are filled with: IPv4 Address: 189.100.0.2, Subnet Mask: 255.255.0.0, Default Gateway: 189.100.0.254, and DNS Server is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	189.100.0.2
Subnet Mask	255.255.0.0
Default Gateway	189.100.0.254
DNS Server	

Figure: PC Configuration of PC5

Ping Communication



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC0. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the output of three ping commands:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 179.100.0.2

Pinging 179.100.0.2 with 32 bytes of data:

Request timed out.
Reply from 179.100.0.2: bytes=32 time=2ms TTL=125
Reply from 179.100.0.2: bytes=32 time=2ms TTL=125
Reply from 179.100.0.2: bytes=32 time=23ms TTL=125

Ping statistics for 179.100.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 23ms, Average = 9ms

C:\>ping 189.100.0.1

Pinging 189.100.0.1 with 32 bytes of data:

Reply from 189.100.0.1: bytes=32 time=3ms TTL=126
Reply from 189.100.0.1: bytes=32 time=13ms TTL=126
Reply from 189.100.0.1: bytes=32 time=1ms TTL=126
Reply from 189.100.0.1: bytes=32 time=1ms TTL=126

Ping statistics for 189.100.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 4ms

C:\>ping 179.100.0.1

Pinging 179.100.0.1 with 32 bytes of data:

Reply from 179.100.0.1: bytes=32 time=16ms TTL=125
Reply from 179.100.0.1: bytes=32 time=1ms TTL=125
Reply from 179.100.0.1: bytes=32 time=7ms TTL=125
Reply from 179.100.0.1: bytes=32 time=1ms TTL=125

Ping statistics for 179.100.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 6ms

C:\>ping 189.100.0.2

Pinging 189.100.0.2 with 32 bytes of data:
```

Figure: Ping Command

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 169.100.0.1

Pinging 169.100.0.1 with 32 bytes of data:

Reply from 169.100.0.1: bytes=32 time=1ms TTL=125
Reply from 169.100.0.1: bytes=32 time=15ms TTL=125
Reply from 169.100.0.1: bytes=32 time=13ms TTL=125
Reply from 169.100.0.1: bytes=32 time=12ms TTL=125

Ping statistics for 169.100.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 10ms

C:\>ping 169.100.0.2

Pinging 169.100.0.2 with 32 bytes of data:

Reply from 169.100.0.2: bytes=32 time=2ms TTL=125
Reply from 169.100.0.2: bytes=32 time=1ms TTL=125
Reply from 169.100.0.2: bytes=32 time=14ms TTL=125
Reply from 169.100.0.2: bytes=32 time=9ms TTL=125

Ping statistics for 169.100.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 6ms

C:\>ping 179.100.0.1

Pinging 179.100.0.1 with 32 bytes of data:

Request timed out.
Reply from 179.100.0.1: bytes=32 time=8ms TTL=126
Reply from 179.100.0.1: bytes=32 time=1ms TTL=126
Reply from 179.100.0.1: bytes=32 time=23ms TTL=126

Ping statistics for 179.100.0.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 23ms, Average = 10ms

C:\>ping 179.100.0.2

Pinging 179.100.0.2 with 32 bytes of data:

```

Figure: Ping Command

Test Results Between PCs

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC3	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC5	PC1	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC4	PC2	ICMP		0.000	N	3	(edit)	(delete)

Figure: Ping Test Results Between PCs

Conclusion

The network has been successfully designed using Class B IP addressing and configured using static routing. All PCs across different networks can fully communicate with each other, confirming the correctness of routing and addressing.